MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY, INTERCONNECT

This specification is approved for use by the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the Interconnect Circuit Card Assembly: 9355770, used-in the Mortar Ballistics Computer (MBC): M23. Throughout this specification this assembly is referenced as "interconnect".

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specification and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection
MXL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-105	Sampling procedures and Tables for Inspection by Attributes

FSC 1220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-454	-	Electronic Equipment, General Requirements for
MIL-STD-1521	-	Technical Reviews and Audits for Systems, Equipments, and Computer Programs
MIL-STD-45662	-	Calibration System Requirements

(Copies of specificatiofis and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications-form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal.

DRAWINGS

9355770	Circuit Card Assembly, Interconnect
11785850	Computer Set, Ballistics, Mortar: M23

PACKAGING. DATA= SHEETS.

9355770	Packaging of Circuit Card Assembly, Interconnect: 9355-770

(Copies of drawings and packaging data sheets required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or-as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Item definition. The interconnect provides the Mortar Ballistic Computer (MBC) with electrical interconnection between the following:

a. Display/processor circuit card assembly.

b. Memory circuit card assembly.

c. Mfodem circuit card assembly.

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d. Audio interface circuit card assembly.

e. Power supply circuit card assembly.

f. Case interconnect circuit card assembly.

The connector contacts are listed in Table I.

TABLE I. Interconnect assembly connector contacts.

Connector	Pin	Signal Name	Pin	Signal Name
P1	1	GND		
P1	2	GND		
P1	3	GND		
P1	4	L130*		
P1	5	LB1*		
P1	6	LB2*		
P1	7	LB3*		
P1	8	LB4*		
P1	9	HB5*		
P1	10	HB1*		
P1	11	HB4*		
PI	12	HB3*		
P1	13	HBO*		
P1	14	HB2*		
P1	15	MSGLED		
P1	16	ELLO		
P1	17	HB6*		
P1	18	ELHI		
P1	19	ONLED		
P1	20	LB5*		
P1	21	LB6*		
“Pl	22	BATPOS		
P1	23	BATPOS		
P1	24	ON*		
P1	25	SEQLED		
P1	26	BATLED		
XA1	Al	+ 5V	%1	GND
XA1	A2	ABoo	B2	DBO
XA1	A3	ABO1	B3	DB 1
XA1	A4	AB02	B4	DB 2
XA1	A5	AB03	B5	DB 3
XA1	A6	AB04	B6	DB4
XAl	A7	AB05	B7	DB 5
XA1	A8	AB06	B8	DB 6
XA1	A9	AB07	B9	DB7
XA1	Al	AB08	B10	~ *
XA1	All	AB09	Bll	WT*
XA1	A12	ABlo	B12	10/M*
XA1	A13	AB1l	B13	so

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TABLE I. Interconnect assembly connector contracts.- continued

XA1	A14	AB12	B14	S1
XA1	A15	AB13	B15	RESET*
XAl	A16	AB14	B16	LE
XA1	A17	AB15	B17	ALE
XA1	A18	+5V	B18	GND
XA1	A19	PCLK	B19	PCLK
XA1	A20	KEYINT*	B20	KEYINT*
XA1	A21	RTCINT*	B21	RTCINT*
XA1	A22	MODINT*	B22	Spare
XA1	A23	DEMODINT*	B23	RAM+5V
XA1	A24	PSFAULT*	B24	WAIT*
xAl	A25	BREQ*	B25	BACK*
XAl	A26	LBO*	B26	HBO*
XA1	A27	LB1*	B27	HB1*
XA1	A28	LB2*	B28	HB2*
XA1	A29	LB3*	B29	HB3*
XA1	A30	LB4*	B30	HB4*
XA1	A31	LB5*	B31	HB5*
XA1	A32	LB6*	B32	HB6*
XA1	A33	+5V	B33	GND
XA1	A34	Spare	B34	614.4KHZ
XA1	A35	Spare	B35	5HZ
XA1	A36		B36	2HZ
XA1	A37		B37	1HZ
XA1	A38		B38	2SEC
XA1	A39		B39	3SEC
XA1	A40	WAIT*	B40	6SEC
XA1	A41	OFF	B41	LEDDC
XA1	A42	ELON	B42	DSPON*
XA1	A43	ANDAT	B43	ANDAT
XAl	A44	ANCLK	B44	ANCLK
XA1	A45	GRI)EN	B45	GRDEN
XA1	A46	GRDCLK	B46	GRDCLK
XA1	A47	EOF	B47	EOF
XA1	A48	320Hz	B48	320Hz
XA1	A49		B49	+35V
XA1	A50	-5V	B50	GND
XA2	Al	+ 5V	B1	GND
XA2	A2	ABOO	B2	DB 0
XA2	A3	ABO1	B3	DB 1
XA2	A4	AB02	B4	DB 2
XA2	A5	AB03	B5	DB 3
XA2	A6	AB04	B6	DB 4
XA2	A7	AB05	B7	DB 5
XA2	A8	AB06	B8	DB 6
XA2	A9	AB07	B9	DB 7
XA2	A10	AB08	Blo	RD*
XA2	All	AB09	Bll	WT*
XA2	A12	AB10	B12	10/M*

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TABLE I. Interconnect assembly connector contracts. Continued

XA2	Al 3	ABll	B13	S0
XA2	A14	AB12	B14	S1
XA2	Al 5	AB13	B15	Spare
XA2	A16	AB14	B16	LE
XA2	Al 7	AB15	B17	ALE
XA2	Al 8	+5V	B18	GND
XA2	A19	Spare	B19	PCLK
XA2	A20		B20	Spare
XA2	A21		B21	Spare
XA2	A22		B22	BAT+5V
XA2	A23		B23	RAM+5V
XA2	A24	Spare	B24	Spare
XA2	A25	BREQ*	B25	BACK*
XA2	A26	Spare	B26	Spare
XA2	A27		B27	
XA2	A28		B28	
XA2	A29		B29	
XA2	A30		B30	
XA2	A31		B31	
XA2	A32	Spare	B32	Spare
XA2	A33	+5V	B33	GND
XA2	A34	Spare	1334	Spare
XA2	A35	Spare	B35	
XA2	A36		B36	
XA2	A37		B37	
XA2	A38		B38	
XA2	A39		B39	
XA2	A40		B40	
XA2	A41		B41	
XA2	A42		B42	
XA2	A43		B43	
XA2	A44		B44	
XA2	A45		B45	
XA2	A46		B46	
XA2	A47		B47	
XA2	A48		B48	
XA2	A49	Spare	B49	Spare
XA2	A50	-5V	B50	GND
XA3	Al	+5V	B1	GND
XA3	A2	AB00	B2	DBO
XA3	A3	AB0l	B3	DB 1
XA3	A4	AB02	B4	DB2
XA3	A5	AB03	B5	DB 3
XA3	A6	AB04	B6	DB4
XA3	A7	AB05	B7	DB 5
XA3	A8	AB06	B8	DB6
XA3	A9	AB07	B9	DB7
XA3	A10	AB08	B10	RD*
XA3	All	AB09	Bll	WT*

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TABLE I. Interconnect assembly connector contacts.-Continued

XA3	Al2	AB1O	B12	10/M*
XA3	A13	AB1l	B13	so
XA3	A14	AB12	B14	S1
XA3	A15	AB13	B15	Spare
XA3	A16	AB14	B16	LE
XA3	A17	AB15	B17	ALE
XA3	A18	+5V	B18	GND
XA3	Al 9	CFSK	B19	PCLK
XA3	A2 0	WXMT	B20	WIREN
XA3	A21	MFSK	B21	Spare
XA3	A22	MODINT*	B22	XMTKY
XA3	A23	DEMODINT*	B23	XMTKYR
XA3	A24	MC600	B24	MC600
XA3	A25	BREQ*	B25	BACK*
XA3	A26	DACLK*	B26	DACLK*
XA3	A27	ROMICLK*	B27	ROMICLK*
XA3	A28	ROM2CLK*	B28	ROM2CLK*
XA3	A29	RCDATA	B29	RCDATA
XA3	A3 0	RCCLK	B30	RCCLK
XA3	A31	DCCFSK	B31	DCCFSK
XA3	A32	MC19K	B32	MC19K
XA3	A33	+5V	B33	GND
XA3	A34		B34	614.4HKHz
XA3	A35		B35	5HZ
XA3	A36		B36	2EZ
XA3	A37		B37	lHZ
XA3	A38		B38	2SEC
XA3	A39		B39	3SEC
XA3	A4 0		B40	6SEC
XA3	A41		B41	LEDDC
XA3	A42		B42	DSPON*
XA3	A43	Spare	B43	BATLOW
XA3	A44	Spare	B44	Spare
XA3	A45	SEQLED	B45	OI?LED
XA3	A46	BATLED	B46	MSGLED
XA3	A47	MC9.6K	B47	MC9.6K
XA3	A48	MC2400	B48	MC2400
XA3	A49	MC1200	B49	MC1200
XA3	AS 0	-5V	B50	GND
XA6A1A2	1	CFSK		
XA6AlA2	2	WXMT		
XA6AlA2	3	WIREN		
XA6A1A2	4	MFSK		
XA6AlA2	5	+5V		
XA6AlA2	6	-5V		
XA6AlA2	7	GND		
xA2AlA2	8	Spare		
XA6AlA2	9	XMTKYR		
XA6AlA2	10	XMTKY		

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TABLE I. Interconnect assembly connector contracts.-Continued

XA6A1PS1A1	1	BAT+5V
XA6A1PS1A1	2	- 5V
XA6A1PS1A1	3	GND
XA6A1PS1A1	4	GND
XA6A1PS1A1	5	BATPOS
XA6A1PS1A1	6	BATPOS
XA6A1PS1A1	7	+5V
XA6A1PS1A1	8	+5V
XA6A1PS1A1	9	GND
XA6A1PS1A1	10	GND
XA6A1PS1A1	11	GND
xA6AlPSlAl	12	RESET*
XA6A1PS1A1	13	PSFAULT*
XA6AiPSlAl	14	BATLOW
XA6A1PS1A1	15	0E'F
XA6A1PS1A1	16	ELHI
XA6A1PS1A1	17	ON*
XA6A1PS1A1	18	ELON
XA6A1PS1A1	19	ELLO
XA6A1PS1A1	20	DSPON*
XA6A1PS1A1	21	+35V
XA6A1PS1A1	22	MC1200

3.2 Performance. The performance of the interconnect shall be in accordance with the following requirements.

3.2.1 Functional. The resistance between the connector contacts listing in Table II shall be less than 7.5 Ohms.

TABLE II. Connector contacts; resistance.

P1-l	to	P1-2	to	P1-3	to	XA1-B1	to
XA1-B18	to	XA1-B33	to	XA1-B50	to	XA2-B1	to
XA2 -B18	to	XA2-B33	to	XA2-B50	to	XA3-B1	to
XA3-B18	to	XA3-B33	to	XA3-50	to		
XA6AlA2-7	to	XA6A1PS1A1-3	to	XA6A1PS1A1-4	to		
XA6A1PS1A1-9	to	XA6A1PS1A1-10	to	XA6A1PS1A1-11			

P1-4	to	XA1-A26
P1-5	to	xAl"A27
P1-6	to	XA1-A28
P1-7	to	XA1-A29
P1-8	to	XA1-A30
P1-9	to	XA1-B31
P1-lo	to	XA1-B27
P1-lo	to	XA1-B30
P1-12	to	XA1-B29
P1-13	to	XA1-B26
P1-14	to	XA1-B28

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TABLE II. Connector contacts; resistance.-Continued

P1-15	to	XA3 -B46					
P1-16	to	XA6A1PS1A1-19					
P1-17	to	XA1-B32					
P1-18	to	XA6A1PS1A1-16					
P1-19	to	XA3-B45					
P1-20	to	XA1-A31					
P1-21	to	xA1-A32					
P1-22	to	Pi-23	to	XA6A1PS1A1-5	to	XA6A1PS1A1-6	
P1-24	to	XA6A1PS1A1-17					
P1-26	to	XA3-A45					
P1-26	to	XA3-A46					
XA1-A1	to	XA1″A18	to	XA1-A33	to	XA2-A1	to
XA2-A18	to	XA2-A33	to	XA3-A1	to	XA3-A18	to
XA3-A33	to	XA6A1A2-5	to	XA6A1PS1A1-7	to	XA6A1PS1A1-8	
XA1-A2	to	XA2 -A2	to	XA3 -A2			
xA1-A3	to	XA2-A3	to	XA3-A3			
XA1-A4	to	XA2 -A4	to	XA3-A4			
XA1-A5	to	XA2-A5	to	XA3-A5			
XA1-A6	to	XA2 -A6	to	XA3-A6			
XA1-A7	to	XA2-A7	to	XA3-A7			
XA1-A8	to	XA2-A8	to	XA3 -A8			
XA1-A9	to	XA2-A9	to	XA3-A9			
XA1-A10	to	XA2 -A10	to	XA3-A10			
XA1-All	to	XA2-A1l	to	XA3-A1l			
XA1-A12	to	XA2-A12	to	XA3-A12			
XA1-A13	to	XA2-A13	to	XA3-A13			
XA1-A14	to	XA2-A14	to	XA3-A14			
XA1-A15	to	XA2-A15	to	XA3-A15			
XA1 -A16	to	XA2 -Al 6	to	XA3-A16			
XA1 -A17	to	XA2-A17	to	xA3-A17			
XA1-A19	to	XA1-B19	to	XA2-B19	to	XA3-B19	
XA1-A20	to	XA1-B20					
XA1-A21	to	XA1-B21					
xA1-A22	to	XA3-A22					
XA1-A23	to	XA3-A23					
XA1-A24	to	XA6A1PS1A1-13					
XA1-A25	to	XA2-A25	to	XA3-A25			
XA1-A40	to	XA1-B24					
XA1-A41	to	XA6A1PS1A1-15					
XA1-A42	to	xA6AlPslA1-18					
XA1-A43	to	XA1-B43					
xA1-A44	to	xA1-B44					
xA1-A45	to	XA1-B45					
XA1-A46	to	XA1-B46					
xA1-A47	to	XA1-B47					
XA1-A48	to	XA1-B48					

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TABLE II. Connector contacts; resistance.-Continued

XA1-A50	to	XA2-A50	to	XA3-A50	to	XA6A1A2-6	to XA6A1PS1A1-2
XA1-B2	to	XA2-B2	to	XA3-B2			
XA1-B3	to	XA2-B3	to	XA3-B3			
XA1-B4	to	XA2-B4	to	XA3-B4			
XA1-B5	to	XA2-B5	to	XA3-B5			
XA1-B6	to	XA2-B6	to	XA3-B6			
XA1-B7	to	XA2-B7	to	XA3-B7			
XA1-B8	to	XA2-B8	to	XA3-B8			
XA1-B9	to	XA2-B9	to	XA3-B9			
XA1-B10	to	XA2-B10	to	XA3-B10			
XA1-B1l	to	XA2-B1l	to	XA3-B1l			
XA1-B12	to	XA2-B12	to	XA3-B12			
XA1-B13	to	XA2-B13	to	XA3-B13			
XA1--B14	to	XA2-B14	to	XA3-B14			
XA1-B15	to	XA6A1PS1A1-12					
XA1-B 16	to	XA2-B16	to	XA3-B16			
xA1-B17	to	XA2-B17	to	XA3-B17			
XA1-B23	to	XA2-B23					
XA1-B25	to	XA2-B25	to	xA3-B25			
XA1-B34	to	XA3-B34					
XA1-B35	to	XA3-B35					
XA1-B36	to	XA3-B36					
XA1 -B37	to	XA3-B37					
XA1-B38	to	XA3-B38					
XA1-B39	to	XA3-B39					
XA1-B40	to	XA3-B40					
XA1-B41	to	XA3-B41					
XA1-B42	to	XA3-B42	to	XA6A1PS1A1-20			
XA1-B49	to	XA6AlPSlA1-21					
XA2 -B22	to	XA6A1PS1A1-1					
XA3-A19	to	XA6A1A2-1					
XA3-A20	to	XA6A1A2-2					
XA3-A21	to	XA6AIA2-4					
XA3 -AZ 4	to	XA3-B24					
XA3-A26	to	XA3-B26					
XA3-A27	to	XA3-B27					
XA3-A28	to	XA3-B28					
XA3-B29	to	XA3-B29					
XA3 -B3 0	to	XA3-B 30					
XA3-B31	to	XA3-B31					
XA3 -B32	to	XA3-B32					
XA3-B47	to	XA3-B47					
XA3-B48	to	XA3-B48					
XA3-B49	to	XA3-B49	to	XA6A1PS1A1-22			
XA3 -B 20	to	XA6A1A2-3					
XA3-B22	to	XA6A1A2-10					
XA3 -B23	to	XA6AlA2-9					
XA3-B43	to	zA6A1PS1A1-14					

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3.2.2 Physical characteristics. The physcial characteristics shall be as prescribed by the drawings, specifications, and QAP's.

3.2.3 Reliability. When specified in the contract, a special sample shall be subjected to reliability testing. The manufacturing and production techniques employed in the fabrication of the interconnect shall not degrade the inherent reliability. The inherent reliability shall not be less than 2,449,479 hours Mean Time Between Failures (MTBF). (See 4 and 6)

Environmental. The interconnect shall meet the requirements specified herein unless otherwise specified.

3.3.1 Room ambient temperature. The requirements of this specification shall be met with the interconnect thermally stabilized at room ambient temperature of +16°C to +32°C [+60°F to +90°F), unless otherwise specified in the requirements paragraph.

3.3.2 Environmental stress screen (ESS) temperature. The interconnect shall meet the requirements of 3.2.1 after having been exposed to a minimum of ten temperature cycles at temperatures of -57 C and +71°C (-71°F and +160°F).

3.3.3 Environmental stress screen (ESS) vibration. The interconnect shall meet the requirements of 3.2.1 after having been subjected to a random or pseudorandom vibration which includes frequency content from 100 Hz to 1000 Hz and up to a 3g RMS in two axis for a period of 10 minutes plus or minus 15 seconds.

3.4 Fabrication. The interconnect shall be manufactured in accordance with drawing 9355770.

3.5 First article. When specified in the contract, a Sample shall be subjected to first article inspection (See 4 and 6) .

Workmanship. The workmanship requirements of MIL-F-13926 and MIL-STD-454 shall be met. All components shall be fitted properly and firmly in their respective positions. All electrical connections shall be mechanically secure and electrically sound. There shall be no evidence of corrosion, burrs, roughness, sharp edges or other defects which could adversely affect its use or suitability for the purpose intended-

3.7 Materials. Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is

specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 and MIL-STD-454 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval test shall consist of three (3) interconnects plus three (3) each of all items covered by QAP. The three interconnects shall be tested as specified herein for all the requirements of this specification. The items covered by QAP shall be tested, as specified thereon, for all characteristics delineated on the QAP. The samples shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as will be used in the production of the balance of the contract quantity. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in the production of the balance of the contract quantity.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for all materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for all materials, shall be submitted to the Government for approval. The Government reserves the right to witness any or all of the contractor’s inspections.

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4.3 Inspection provisions.

4.3.1 Submission of product. unless otherwise specified by the contracting officer, inspection lot size, lot formation, and presentation of lots shall be in accordance with "Submission of product" provision of MIL-STD-105.

4..3.2 Examination and tests.

4.3.2.1 Components and subassemblies, all components and subassemblies shall be inspected in accordance with the inspection provisions contained in Quality Assurance Provisions (QAP'S listed in the Technical Data Package (TDP). In the absence of QAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4..4 Sampling.

4.4.1 Environmental\performance. The interconnect shall be inspected to the requirements in Table III.

TABLE III. Classification of defects.

No.	Characteristics	Requirement	Test procedure
100% Inspection			
1.	Functional test	3.2	4.6.3
2.	ESS temperature	3.3.2	4.6.1
3.	ESS Vibration	3.3.3	4.6.2
MAJOR: AQL 1.0% Defective			
101.	Workmanship	3.6	4.6.4

4.4.2 Special sampling. one interconnect shall be selected at random as a special sample from each lot or each 100 produced which ever occurs first. The sample shall meet the requirements and tests in Table IV.

TABLE IV. Classification of defects.

No.	Characteristics	Requirement	Test procedure
201.	Fabrication	3.4	Applicable Drawing/ Visual
202.	Materials	3.7	Applicable Drawing/ Visual

NOTE: The tests in Table III shall be conducted at room ambient temperature (as specified in 3.3.1).

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4.4.3 Failure of sample. Should any one item of a special sampling fall to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and resubmitted items have been approved in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.5 Inspection equipment. Except as otherwise provided for bv the contract, the contractor shall suPPlY and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-1-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. All inspection equipment specified by drawing number in specification or QAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the Technical Data Package List (TDPL).

4.5.2.2 Contractor design. In the absence of Government inspection equipment design, the contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolance used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Temperature testing. The volume of the test chamber shall conform to the test facilities requirements of MIL-F-13926. The air temperature at the control sensor shall be in accordance with the specified requirements within plus or minus 2.5°F (1.4°C). The temperature gradient across the cross-sectional area occupied by the test item shall not exceed

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0.5°F per foot in any direction. The heat source of the test chamber shall be so located that radient heat, if used, will not fall directly on the test item. Theromocouples or equivalent temperature sensors utilized to determine or control the specified chamber temperature shall be centrally located within the test chamber within the supply airstream, or in the return airstream, whichever provides the specified test condition at the item under test and shall be baffled or otherwise protected against radiation effects.

4.5.2.3.2 Fabrication, setup and calibration. Design and fabrication of test equipment shall be such that it incorporates the features of rigidity, stability, and ease of maintenance. Setup and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment. Operating and calibration instruction shall be prepared and submitted to the procuring activity.

4.6 Test methods and procedures. Unless otherwise specified in the contract, the following procedures shall be utilized to conduct specified tests. If the contractor chooses to use alternate test procedures, a detailed description of the set-up and procedure shall be submitted to the government procuring agency for approval prior to use.

4.6.1 ESS temperature. Testing equipment utilized in this test shall conform to the requirements of 4.5.2.3.1. The interconnect shall be placed in the test chamber and the temperature of the chamber reduced 10oC (180F) per minute plus or minus 1°C (1.8oE') from room ambient temperature (as specified in 3.3.1) to -57°C (-71°F). The interconnect shall be maintained at this temperature for a minimum of 20 minutes. Upon completion of the 20 minute exposure, the temperature of the chamber shall be raised 100C (18°F) per minute plus or minus 1°C (1.8°F) to +71°F (+160°C). The interconnect shall remain at this temperature for a minimum of 20 minutes. Upon completion of the 20 minute exposure the temperature of the chamber shall be reduced 10°C (180F) per minute plus or minus 10C (1.80F) to room ambient temperature (as specified in 3.3.1). The above procedure shall be conducted a minimum of ten times. Upon completion of the cycles the interconnect shall then be removed from the chamber and subjected to the test in 4.6.3. It shall meet the requirements of 3.2.1.

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4.6.2 ESS vibration. Conduct the vibration test with the interconnect positioned on an appropriate adapter affixed to a vibrating machine that is capable of providing the amplitudes and frequencies specified in 3.3.3. unless otherwise specified, the test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The applied motion shall be in two separate planes for the time period specified. Upon completion of the vibration sweep the interconnect shall be removed from the vibration adapter and shall be subjected to the test in 4.6.3. It shall meet the requirements of 3.2.1.

4.6.3 Functional tests. This test is to be conducted utilizng a certified Ohm meter. Measure the resistance between the connector contacts listed in Table II. All measures resistances shall meet the requirements of 3.2.1.

4.6.4 Workmanship. The interconnect shall be subjected to a visual and tactile examination for evidence of faulty workmanship. The assembly shall conform to the requirements of 3.6.

4.7 Reliability testing. The requirement for reliability test approval and the responsibility (Government or contractor) for reliability testing shall be as specified in the contract. The sample for reliability test shall consist of 3 interconnects. The samples shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as will be used in the production of the balance of the contract quantity. All parts and materials, including packaging and packing shall be obtained from the same source of supply as will be used in the production of the balance of the contract quantity.

4.7.1 Government testing. When the Government is responsible for conducting reliability tests, the contractor prior to submitting the sample to the Government, shall inspect the sample to assure that it conforms to all requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for all materials.

4.7.2 Contractor testing. When the contractor is responsible for conducting reliability tests, the sample shall be inspected and tested by the contractor for all the requirements of the contract. The sample and a record of these inspections and tests, including certificates of conformance for all materials, shall be submitted to the Government for approval.The Government reserves the right to witness any or all of the contractors reliability tests and inspections.

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5. PREPARATION FOR DELIVERY

5.1 Delevery requirements. Handling, packaging, packing, shipping, and marking of the interconnect, 9355770 shall be in accordance with Packaging Data Sheet 9355770. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The interconnect described herein is intended for use In the MBC. See drawing 11785850.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Provisions for first article testing.

c. Provisions for reliability testing.

d. Applicable Packaging Data Sheet.

Applicable levels of preservation, packing, and markinq (see 5.1).

f. Applicable stock numbers.

Custodian:
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Preparing activity:
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(Project 1220-A309)

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-70498 (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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